

PORTFOLIO CASE STUDY

E-Commerce Business Intelligence & Analytics Pipeline

From Raw Data to Revenue-Saving Decisions

UK Online Retail Dataset · 2010 – 2011 · ~540,000 Transactions Analyzed



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The Story Behind the Data

An online retailer had roughly 540,000 transactions sitting untouched in spreadsheets.

Leadership could see the numbers, but couldn't turn them into decisions — and three simple questions kept going unanswered.



THE GOAL

Transform this messy data into a clear, actionable dashboard that answers leadership's toughest questions in seconds.



Who are our most valuable customers, and are they leaving us?



Why is customer retention falling year over year?



What products should we bundle to increase revenue?

How I Built It — From Data to Dashboard

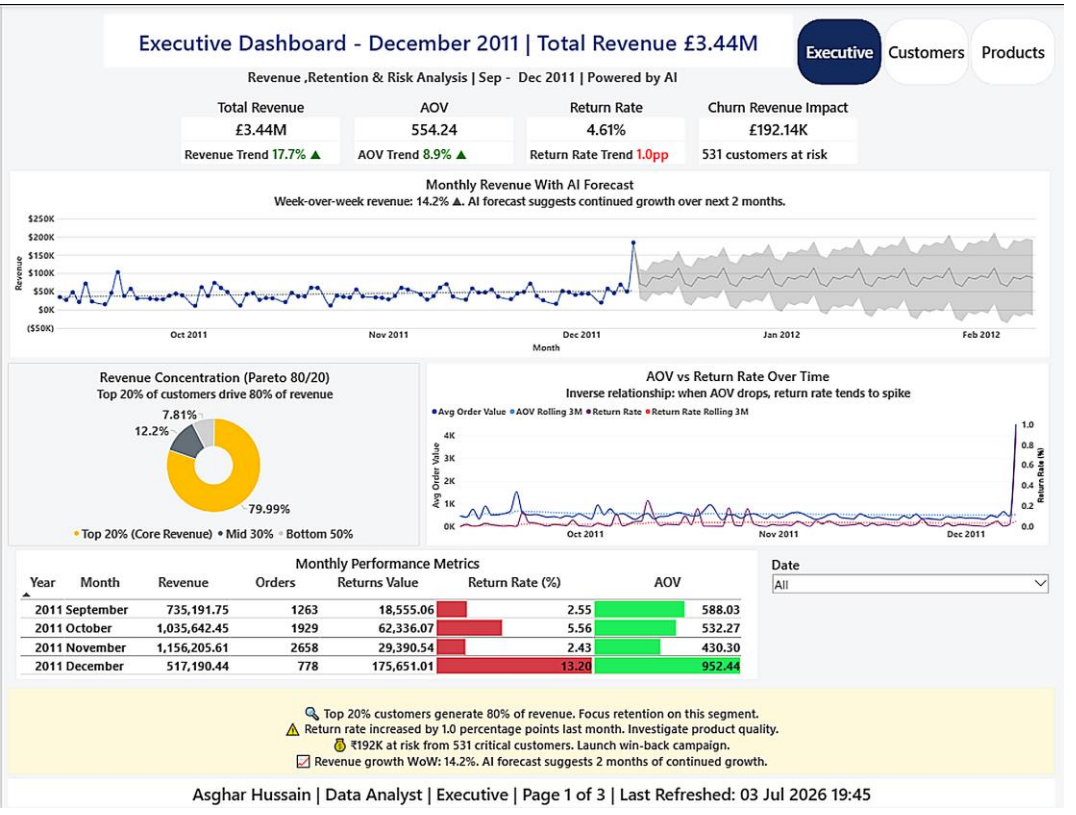


WHY THIS STACK

Excel for quick exploration, SQL for heavy logic, and Power BI for storytelling — each tool doing what it does best.



Business Health at a Glance



£3.44M
Total Revenue · +17.7% trend



£554.24
Average Order Value · +8.9%



4.61%
Return Rate · +1.0pp last month



£192.14K
Revenue at Risk · 531 customers



80%
Of revenue from top 20% customers



+14.2%
Week-over-week revenue growth



MY TAKE
Protect the top 20% of customers — they are the engine driving this business forward.

Who's Leaving, and Why



Retention Sharply Declined

Twelve-month retention fell from 50.3% for the 2010 cohort to just 12% for 2011 — a drop worth investigating.



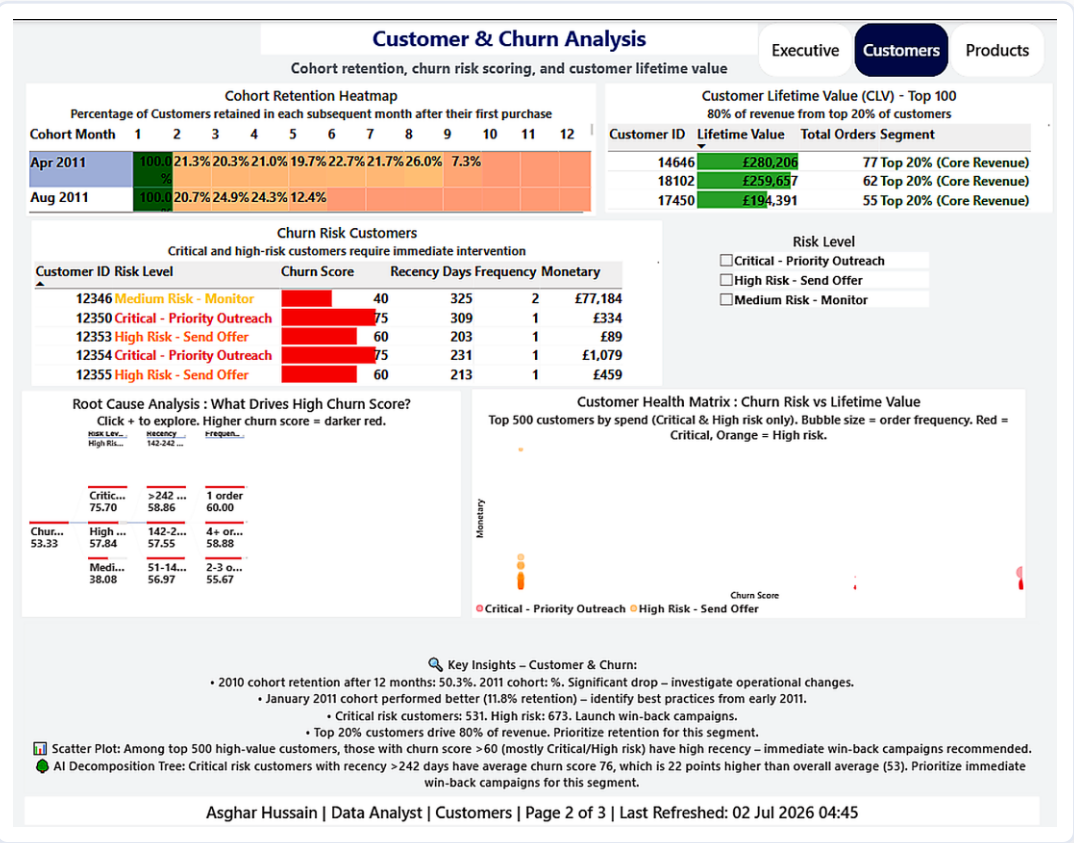
1,204 Customers at Risk

531 flagged Critical and 673 High-Risk by recency, frequency and monetary value — real revenue exposure.



AI Root-Cause Signal

Critical customers inactive 242+ days average a churn score of 76 — 22 points above the 53 overall average.



RECOMMENDED ACTION

Target the “Old & Critical” segment — high churn score, long recency — with personalized win-back offers.

What Sells Together



8 High-Affinity Pairs Found

Of 50 evaluated product pairs, 8 share a lift score of 29.24 — far above the lift-of-1 threshold for real affinity.



One Product Anchors Them All

Product “DOT” appears in every high-lift pair, with 100% purchase confidence each time.



Top Examples

84923 → DOT 15056bl → DOT 47566b → DOT



STRATEGY

Bundle “DOT” with these high-affinity products and offer a 5–10% discount to lift average order value.

What I'd Do Next



VIP Loyalty Program

Protect and reward the top 20% of customers who already generate 80% of total revenue.



Win-Back Campaigns

Launch targeted outreach to the 1,204 Critical and High-Risk customers before they churn for good.



Bundle “DOT” Cross-Sell

Turn the highest-lift product affinity into a real promotional bundle and email campaign.



Investigate the 2011 Retention Drop

Dig into what changed operationally between the 2010 and 2011 cohorts to reverse the decline.

THANK YOU



This project proves that data isn't just numbers. It's a story about your customers — and when you listen to that story, you can save revenue, retain loyalty, and grow smarter.

LET'S CONNECT



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